

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)			
Course Code: MCN301			
Course Name: DISASTER MANAGEMENT			
Max. Marks: 100			Duration: 3 Hours
PART A			
		(Answer all questions; each question carries 3 marks)	Marks
1		Explain the relevance and adverse effects of greenhouse gases. If the entire energy that is received from the sun retained in its earth's surface, the planet would be very hot and would become an inhabitable place. The earth, after heating up its surface, reflects a certain amount of energy. Some of this heat energy is transmitted to the upper layers of air through conduction. The heat energy emitted from the earth's surface is in the form of long wave radiation, and is called outgoing radiation. While a portion of the outgoing radiation is absorbed by certain gases in the atmosphere and retained as heat energy and the remaining energy escapes into outer space. Gases capable of absorbing outgoing radiation are carbon dioxide, carbon monoxide, water vapour, etc. They are called GreenHouse Gases (GHG). Due to the effect of Greenhouse gases, Earth is prevented from cooling down drastically. GHGS thus act like a blanket and provide earth with an ideal climate for life to flourish. This is known as the Greenhouse effect. The intensity of Greenhouse effect varies from place to place depending upon the concentration of GHGS. For instance, the quantity of vapour-and carbon dioxide is less in dry places like deserts. The usage of carbon dioxide and the release of oxygen is high in places like forests where trees are in abundance.	(3)
2		Discuss the two types of monsoon in the Indian subcontinent. <u>The Indian Monsoon</u> Monsoon is a regional wind that blows towards land at a certain season and blows from the landmasses during other seasons. These winds blow in the opposite direction in summer and winter. Though monsoon winds blow over all parts of the world, it is well-developed over India and the South-east Asian regions. The Indian subcontinent has two types of winds.	(3)

	1. South-West Monsoon 2. North-East Monsoon South-West Monsoon The south-east trade winds originate from the southern hemisphere in the Indian Ocean. When these winds cross the equator, they get deflected towards the right by the Coriolis force, becoming the south-west trade winds. These winds gather large quantities of moisture as they pass over the Indian Ocean. As the SW monsoon winds approach the Indian Peninsula, they are diverted into two- the Arabian Sea Branch and the Bay of Bengal Branch. When the moisture laden Arabian Sea branch reaches the south-western side of India, they are blocked by the Western Ghats. When the mountain range blocks the horizontal flow, the wind ascends along the slope of the mountain range, gets cooled down and forms clouds. These clouds then result in precipitation. Kerala gets the south-west monsoon mostly during early June every year. These winds then take a west turn and continue their journey, and spread over the northern parts of India bringing in rains to these areas. Monsoon winds normally reach Delhi in the first week of July and could last till end September/early October. North-East Monsoon Since North-East winds originate mainly from the land masses of the north-east region of India, they are relatively dry. When these winds pass over the Bay Bengal towards south, they gather moisture and cause rainfalls over parts of Odisha, Andhra Pradesh and Tamil Nadu. Cyclone formation is common over Bay of Bengal during the north-east monsoon season. The cyclones also bring in abundant rainfall over Odisha, Andhra Pradesh, Telangana and Tamil Nadu.	
3	State the major data requirements of hazard mapping and the sources for obtaining these data.	(3)

4	State the principle of qualitative risk assessment and the method of expressing risk qualitatively.	(3)
5	State the different types of disaster response. Disaster responses are the set of activities taken during a disaster or immediate following a disaster, directed towards saving life and protecting property. The different types of disaster response include 1. Search and rescue 2. First aid and emergency medical care 3. Evacuation 4. Evacuation centre management 5. Development of Standard Operation Procedure (SOPs) 6. Immediate repair of community facilities and services 7. Relief delivery 8. Coordination and Communication 9. Psycho-social counselling and stress debriefing 10. Medical services	(3)
6	List six international relief organizations. 1. Catholic Relief Services (CRS – USCC) 2. Emergency Nutrition Network (ENN) 3. Doctors Without Borders 4. Food For The Hungry International (FHI) 5. International Federation of Red Cross and Red Crescent Societies (IFRC) 6. International Organisation for Migration (IOM)	(3)
7	Distinguish between risk communication and crisis communication. For public health emergencies, risk communication includes the range of communication capacities required through the preparedness, response and recovery phases of a serious public health event to encourage informed decision making, positive behaviour change and the maintenance of trust. Crisis communication is a sub-specialty of the public relations profession that is designed to protect and defend an individual, company, or organization facing a public challenge to its reputation.	(3)
8	List the structural and nonstructural measures in capacity building. Structural measures undertake to strengthen buildings, lifelines and infrastructure to withstand any hazard. Non-Structural Measure to reduce loss or damage by administrative measures. It does not control or affect the process of inundation.	(3)
9	State the legislation in India on disaster management.	(3)
10	Explain the interrelation of National Disaster Management Policy with other 3 national policies.	(3)
PART B		
(Answer one full question from each module, each question carries 14 marks)		
Module -1		

11	a) Categorize the various layers of atmosphere based on their distance from earth and explain the features of each layer with a neat diagram. Based on chemical composition, the atmosphere is divided into two. Homosphere The Homosphere Extends to a distance of about 88Km from the surface of earth. It consists of gases like nitrogen, oxygen, argon and carbon dioxide. This layer is homogeneous in composition. Heterosphere The atmosphere lies above the homosphere. The air composition is not uniform. The lowermost layer has the heaviest molecules and the upper layer has the lightest elements. Based on the gases present, the heterosphere is further divided to four. Molecular Nitrogen Layer(88Km-200Km) Oxygen Layer (200Km-1125Km) Helium Layer (1125Km-3540Km) Hydrogen Layer (3540Km-9660Km) Based on the temperature, the atmosphere is divided into four parts. Troposphere The bottom dense part, containing 70 percent of the mass, close to the ground is the troposphere. It reaches up to 11 km from the ground. Clouds, storms, fog and haze are found only in the troposphere. The temperature in this layer decreases at about -6.4°C/km with height. This decrease of temperature with altitude is called lapse rate. The border of the troposphere is called Tropopause. Tropopause acts like a lid over the troposphere. Temperature stops decreasing with height from tropopause. Stratosphere The Stratosphere is a clear layer above the troposphere that extends to a height of about 50 km from earth's surface. This layer does not have clouds, storms or dust. Clouds are not formed since water vapors are absent in these regions. Ozonosphere is an important layer found within the stratosphere. Ozone layer is found in this layer. Ozone absorbs and prevents the harmful ultraviolet	(8)
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	radiations from reaching earth, thereby protecting life on earth. The maximum concentration of ozone occurs at 22 km from the ground level. Above the stratosphere, there is a small layer called stratopause where temperature neither increases or decreases with height. 3. Mesosphere The portion of the atmosphere above the stratosphere, between 50 km and 80 km is known as mesosphere. It starts from the edge of Stratopause. Though the temperature in the mesosphere near the stratosphere is higher by about 10°, it falls to -75°C at 80 km. The density of air at this height is about 1/1000 as that of sea level. Mesosphere plays a crucial role in radio communication as ionisation occurs here. The sunlight passing through this layer converts individual molecules to charged ions. These ionised particles are concentrated as a zone in this layer, which is named D-layer. The D-layer reflects radio waves transmitted from earth. Just above the mesosphere is a small layer called Mesopause, where temperature is stable. 4. Thermosphere Thermosphere extends from 80 km to about 60,000 km from earth. Here the temperature increases to about 2000°C. The property of the thermosphere is radically different from the others. Ions are abundant in the thermosphere. In the thermosphere most of the approaching meteoroids burn up before reaching earth. 5. Exosphere The region beyond the thermosphere is called the exosphere. It consists of only hydrogen and helium atoms. This region has very high temperatures due to solar radiation. Based on the chemical reaction, the atmosphere is classified into two. 1. Chemosphere A region of the upper atmosphere between altitudes of 40 and 80 km in which chemical processes driven by sunlight are significant. The chemosphere overlaps the upper stratosphere and the mesosphere	
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		2.Ozonosphere In this region intense chemical reactions take place with the help of oxygen. Ozone layer is present in this region.	
	b)	Define the following terms: a) Disaster b) Hazard c) Risk Disaster - A disaster is a serious disruption, occurring over a relatively short time, of the functioning of a community or a society involving widespread human, material, economic or environmental loss and impacts, which exceeds the ability of the affected community or society to cope using its own resources. Hazard - A physical event, phenomenon or activity that has the potential to cause injury or loss of life, property damage, social and economic disruption or environmental degradation. - E.g.: Earthquake, Flood, Epidemic, Tsunami, Cyclone etc. - Each hazard is characterized by its location, intensity, frequency and probability of occurrence. Risk - Risk is the measure of the expected losses due to a hazard event occurring in a given area over a specific time period. - Risk is a function of the probability of a particular hazardous event and the losses each would cause.” - In the technical sense, it is defined through the combination of three terms: hazard, exposure and vulnerability.	(6)
12	a)	State and explain crisis counselling. Identify the necessity of crisis counselling. <u>Crisis counselling</u> Crisis counselling occurs when a client who is destabilised engages the services of a counsellor. The person is unable to cope with events in his/her life and, consequently, may be wracked by destructive feelings of self-doubt, anxiety, or guilt and may be engaging in hurtful behaviours. This crisis needs immediate attention otherwise there is the risk of further personality or behavioural deterioration. <u>Goals of Counselling</u> The goals of crisis counselling can be summarised as,	(8)

	• help the person return to his usual level of functioning; • decrease anxiety; help people who are in crisis recognise and correct behaviours and cognitive distortions. • teach crisis-solving techniques; and, • give more assistance after immediate help is received. • safety: ensures the individual is safe, any risk has been reduced and resources, if available, have been provided. • stability: ensures the individual is stable and has a short-term plan which includes mastery of self and the emergency or disaster situation. • connection: helps connect the individual to formal and informal resources and support. If resources are not readily available, crisis counselling helps the individual pursue potential natural supports/resources.	
b)	Identify the reasons for the depletion of the Ozone layer. Suggest two initiatives which can be implemented at home to prevent this. <u>OZONE LAYER</u> Ozonosphere is an important layer found within the stratosphere. Ozone is found in this layer. Ozone absorbs and prevents the harmful ultraviolet radiations from reaching earth, thereby protecting life. Without the Ozone layer, life would not have been possible on earth. The maximum concentration of ozone occurs at 22 km from the ground level. Depletion of ozone layer Due to human activities the ozone layer is becoming thin. The thinning of this layer is called <i>ozone depletion</i>. The ozone layer is located in the lower part of the stratosphere between 15 km and 35 km. Concentration of ozone is the maximum at about 25-30 km. The level of ozone is maintained at this level by the Ozone-Oxygen Cycle. When ultraviolet radiation that spreads out from the sun strikes the oxygen molecule, it splits the molecule into two individual oxygen atoms. The oxygen atoms, thus produced, combines with Oxygen (O_2) molecule and produce ozone molecule (O_3). This reaction is aided by either	(6)

	Nitrogen or Oxygen, which absorbs the excess energy that is liberated. Ozone thus formed will be split by ultra-violet rays into a molecule of oxygen and an atom of oxygen (O). It is through this repeated circular ozone and oxygen formation that the concentration of ozone is maintained in the stratosphere. The concentration of ozone in the atmosphere is determined by the rate of its formation and destruction in the above manner. Due to severe depletion of ozone in the atmosphere 'ozone holes' are created. Ozone holes, which were discovered in 1985, are overhead areas having less than 220 Dobson Units (DU). The chemistry of ozone depletion by CFCS, BFCS and Nitric oxides are now discussed. <i>Depletion by CFCs and BFCs</i> CFCS and BFCS are stable compounds in the atmosphere that have the property of living longer (50 to 100 years). Due to their long life, they rise up to the stratosphere. Through the action of UV radiation from the Sun on these compounds, Chlorine (Cl) and Bromine (Br) radicals are released. These radicals act as catalysts, and initiate breaking down of ozone molecules. It is estimated that a single such radical of either Cl or Br is capable of breaking down over a lakh of ozone molecules. Due action, Ozone concentration is decreasing at a drastic rate of four percent per decade. As a result of the inherent long life of CFCs and BFCs, they continue to deplete the ozone layer in a recurrent manner. <i>Depletion by Nitric Oxide</i> One molecule of nitric oxide (NO) combines with ozone; it gets oxidised to nitrogen dioxide and Oxygen. This NO₂ combines with another O₃ molecule to become NO₃ (Nitrate) and O₂. The NO₂ and NO₃ then combine to form N₂O₅ (Dinitrogen pentoxide). Even the atomic oxygen(O) readily combines with NO₂ to yield NO₃. Due to this series of actions and reactions, ozone is completely utilised, and thereby depleted. Large quantities of nitrogen are emitted by aircrafts that community decided to withdraw the operation of jet aircrafts that emit oxides of nitrogen. This step has also helped in reducing the depletion of ozone to a very large extent near the stratosphere.	
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		Module -2	
13	a)	Define hazard mapping. Explain the two approaches of hazard mapping.	(8)
	b)	In Kerala for the past 5 years, the average number of road accidents is 44076 6per year and 1 death occurs in every 10 accidents. Considering the population as 3.33 crores, assess the risk of being killed in driving an automobile in terms of societal and individual risk.	(6)
14	a)	Explain the four different types of vulnerability. List any four socio- economic indicators of human capital as a livelihood asset.	(8)
	b)	Outline the two major physical vulnerability assessment approaches.	(6)
		Module -3	
15	a)	Explain the core elements of disaster risk management. The systematic process of using administrative directives, organizations, and operational skills and capacities to implement strategies, policies and improved coping capacities in order to lessen the adverse impacts of hazards and the possibility of disaster. <pre> graph TD subgraph Cycle A["Risk identification/assessment ▪ Hazard analysis & monitoring ▪ Vulnerability analysis ▪ Determination of risk"] B["Recovery ▪ Rehabilitation ▪ Reconstruction ▪ Rescue services"] C["Preparedness ▪ Early warning ▪ Evacuation ▪ Emergency planning"] D["Prevention and mitigation ▪ Land use planning ▪ Land management ▪ (Non-) structural measures"] A --> B B --> C C --> D D --> A end A <--> B B <--> C C <--> D D <--> A E["DISASTER RISK MANAGEMENT"] F["EVENT"] E --- A E --- B E --- C E --- D E -- Green Arrow --> F </pre> Risk identification and assessment: ▪ This involves determining and analysing the potential, origin, characteristics and behaviour of the hazard ▪ Example:- The frequency of occurrence/magnitude of consequences. Application of risk reduction measures in mitigation: ▪ Planning and implementation of structural interventions (e.g. dams, sea defense) or nonstructural measures such as disaster legislation. Disaster preparedness and emergency management: ▪ Activities and measures taken in advance to ensure effective response to the impact of a hazard, including measures related to timely and effective warnings as well as evacuation and emergency planning. Recovery/Reconstruction: ▪ Decisions and actions taken in the post-disaster phase ▪ Having a view to restoring the living conditions of the affected population.	(10)

	b)	State the requirements for effective disaster response. Disaster response is aimed at providing immediate assistance to maintain life, improve health and support the morale of the affected population. It is focused at meeting the basic needs of the people until more permanent and sustainable solutions can be found. Disaster response depends on the adequacy of preparedness prior to the disaster, considering the scope of responses required. Requirement for Effective Response: Information and resources are two (2) essential requirements for effective response without which plans and efforts at responding will not yield the anticipated results. Information: An early warning system provides vital information for effective response operation despite the unpredictability of some disaster events. An effective warning system must be robust to transmit warnings as early as practicable. Information gained from these systems could help in the planning and decision-making as well as inform the general public. Resources form an essential component of disaster response. The need for disaster management organizations to be resource ready cannot be overemphasized considering the untimely occurrence of disasters, which most often is on short notice. The ability to mobilize the needed resources on short notice is most often hampered by many factors. Its effect on systems gives little room for procrastination of actions.	(4)
16	a)	Define the term 'disaster risk reduction'. Explain the measures for disaster risk reduction. Disaster risk reduction is a systematic and continuous analysis and redressing of the causal factors of disasters. It is the first step and key component of disaster (risk) management. Disaster risk reduction efforts and activities are normally specified in a formal document called disaster risk reduction plan prepared by an appropriate entity or authority. Disaster risk reduction involves structural and non-structural measures. Structural measures include the use of physical or engineering solutions (such as ocean wave barriers or earthquake resistant buildings) to avoid disaster or reduce its impacts. Non-structural measures involve the use of policies, laws, education and awareness creation, and practices to avoid or reduce the impacts of disaster.	(8)
	b)	Define 'relief' in the context of disaster management. Identify the principles guiding relief. 'Relief' is defined as the provision of assistance or intervention during or immediately after a disaster to meet the life preservation and basic subsistence needs of those people affected. It can be of an immediate, short-term, protracted duration. Relief, as a disaster management process, provides timely essential needs such as basic household items, shelter, food, water and sanitation, or health items. Relief, globally, is guided by fundamental principles which disaster organisations and NGOs engaging in disaster relief response ought to apply. The principles require that: ☐ Response to disasters must have humanitarian imperative ☐ Aid is provided based on needs alone and must be done without discrimination of any kind (race, creed or nationality of the recipients) ☐ Aid will not be used to further a particular political or religious standpoint ☐ Aid agencies do not to act as instruments of government foreign policy ☐ That culture and custom are respected in response and relief activities ☐ Disaster response is built on local capacities ☐ Ways be are developed to involve programme beneficiaries in the management of relief aid ☐ Relief aid would reduce future vulnerabilities to disaster as well as meeting basic needs ☐ Accountable to both those we seek to assist and those from who we accept resources ☐ Information, publicity and advertising activities shall recognise disaster victims as dignified human beings and not hopeless objects.	(6)
		Module -4	

17	a) Describe the effective ways of promoting stakeholder participation in disaster risk reduction. State its benefits. Stakeholder participation is widely acknowledged as a critical component of post-disaster recovery because it helps create a shared understanding of local hazard risk and vulnerability, improves recovery and mitigation decision efficacy, and builds social capital and local resilience to future disasters. Communities, particularly those most vulnerable are the key stakeholders in disaster management. These are most vital to people-centred early warning systems. Their input into system design and their ability to respond ultimately determine the extent of risk associated with natural hazards. Key issues in context with the communities are as follows: • The vulnerable communities need to be aware of hazards and potential negative impacts to which they are exposed and be able to take specific actions to minimize the threat of loss or damage. • The most essential determinant of the selection of the disasters on which system should focus is the geographical location of such communities. • For example, while the coastal communities need to be educated and prepared for the possibility of a tsunami, a community in Himalayas can be educated to respond to an early warning system for landslides and earthquakes.	(8)
	b) Explain the basic steps in participatory stakeholder engagement. Stakeholder participation is an increasingly accepted component of natural resources and environmental planning processes in the United States and some parts of the world. In the U.S., stakeholder participation has been codified in environmental planning (e.g., the Administrative Procedure Act and the National Environmental Policy Act). Outside of the United States, international bodies such as the European Union, the World Bank, and World Commission on Dams have incorporated stakeholder participation into policy making and planning procedures. The importance of stakeholder participation in international water resources planning increased in the 1970s and continued through the 1990s and into 2000 and beyond (ESMPAP, 2003; IUCN, 2000; Mol, 2001; OECD, 1973; World Commission on Dams, 2000).	(6)

	National Academies of Sciences, Engineering, and Medicine. 2004. Analytical Methods and Approaches for Water Resources Project Planning. Washington, DC: The National Academies Press. https://doi.org/10.17226/10973. Ideally, expert and public input would be integrated throughout the planning process, but one or the other may play a leading role at different times. Although their respective roles may vary through the planning process, useful stakeholder participation will engage decision makers and stakeholders at all planning stages. Key points in the planning process include the following: National Academies of Sciences, Engineering, and Medicine. 2004. Analytical Methods and Approaches for Water Resources Project Planning. Washington, DC: The National Academies Press. https://doi.org/10.17226/10973. Problem formulation. At this stage, key questions are identified for further data gathering and analysis. To capture the complexity of a problem, scientific inputs as well as inputs from the affected parties are needed. Problem formulation should involve stakeholders, including the affected public, interest groups, agency officials, and scientific or technical experts. Process design. This involves establishing key parameters in the planning process, such as the appropriate scale that captures the full impact of a project and the full range of affected stakeholders (see the 216 study report on river basins and coastal systems for more detailed discussion of the issue of appropriate scales of planning). Experts can play important roles in identifying the appropriate scale of analysis by applying scientific theories to answer questions such as the following: What scale is best for addressing the problem or issue at hand? Can managers effectively influence critical areas, given the scale selected? Can the problem and the proposed solution be credibly evaluated? Is management at the selected scale feasible and economically affordable? Selecting options and desired outcomes. This is made possible by identifying trade-offs among alternative solutions and by identifying publicly held values that guide the selection of options. Scientific inputs provide a basis to identify trade-offs between alternatives. To understand the full range of trade-offs and alternatives, a variety of scientific inputs (ecological, physical, engineering, and behavioral) may be needed. Trade-offs may entail conflicts between fundamental and legitimate differences in values. It is especially important to recognize, respect, and deliberate on these points early in the study process.	
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18	a) Explain capacity building, relevance of capacity assessment and the different methods of assessing capacity in disaster risk management. Capacity-building is defined as the process of developing and strengthening the skills, instincts, abilities, processes and resources that organizations and communities need to survive, adapt, and thrive in a fast-changing world. A Capacity Assessment is an analysis of desired capacities against existing capacities; this generates an understanding of capacity assets and needs, which informs the formulation of a capacity development response. Capacity assessment is the process by which the capacity of a group is reviewed against desired goals, where existing capacities are identified for maintenance or strengthening and capacity gaps are identified for further action. Capacity resides at three related levels: in individuals, in organizations and in the overall working environment within which individuals and organizations operate - 'the enabling environment', which strongly relates to the concept of resilience. Each of these can be an entry point for capacity assessment: Enabling environment Sometimes referred to as the 'societal' or 'institutional' level, capacities at the level of the enabling environment relate to the broader system within which individuals and organizations function. Understanding the enabling environment can be obtained from the 'institutional analysis', 'power analysis' or 'drivers of change analysis' increasingly being undertaken by donor organizations as the basis for country assistance plans. Capacities at the level of the enabling environment relate to all the rules, laws and legislation, policies, power relations and social norms. Governments, civil society and the private sector therefore have an opportunity and obligation to work together to commit to a safer future, and therefore their capacity for engagement can be assessed across all sectors (e.g. climate change, finance, planning) and levels (e.g. small and medium enterprise, farmers, insurers). The Organizational Level This level is a common entry point for capacity assessment. This level relates to the internal structure, policies, systems and procedures that determine an organization's effectiveness and ability to deliver on its mandate and allow individuals to work together. Organizational level capacities help develop and apply internal policies,	(10)
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	arrangements, procedures and frameworks, which is necessary to deliver the organization's mandate. Organizational level capacities help develop and apply internal policies, arrangements, procedures and frameworks, which are necessary to deliver the organization's mandate. It is not only about skills, but also incentives and governance. People and organizations can have strong or weak incentives to change, develop and learn, as a result of their environment or internal factors. The Individual Level This level relates to the skills, experience and knowledge of people that allow them to perform. Capacity assessment at this level is commonly implemented by researchers and non-governmental organizations working at the local level, as well as by some local level governments. However, individual capacity has to be understood within the context of both the organizational level and enabling environment. Four key issues common to most capacity assessments are institutional arrangements, leadership, knowledge and accountability. Not every assessment needs to cover all four of these issues, but they should be at least considered when defining the scope of an assessment.	
b)	State the barriers to effective communication in disaster management. Communication problems are pervasive in many disaster situations. Nearly all the case studies of disasters occurring in the last 30 years describe communication problems of some sort. The general issues are: 1) external communication with the public (covered in another post); 2) internal communication among emergency response personnel and their agencies, discussed here.	(4)

		Despite massive technological changes in recent years, communication problems persist: including system failure, system overload, and incompatibility between communication systems used by different agencies.	
		Module -5	
19	a)	Explain the common disaster types in India.	(10)
	b)	State the objectives and main elements of national disaster management policy.	(4)
20	a)	State the targets, priorities and guiding principles of Sendai Framework for disaster risk reduction.	(8)
	b)	Explain the institutional arrangement for disaster management in India	(6)